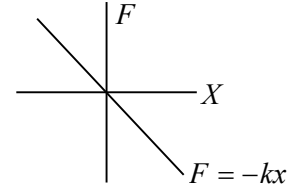


Simple Harmonic Motion

Any periodic motion is identified as simple harmonic motion if motion is periodic and there restoring force acting on a particle of mass m . If restoring force is linear with displacement from equilibrium then motion is identified as Simple harmonic motion. The best example of simple harmonic motion is spring mass system which is also identified as simple harmonic oscillator. The force corresponds to harmonic oscillator is shown in figure.



Equation of motion of simple harmonic oscillator $m \frac{d^2x}{dt^2} = -kx$

The equation reduces to $\frac{d^2x}{dt^2} + \omega^2 x = 0$ where $\omega = \sqrt{\frac{k}{m}}$ is identified as angular frequency of system for harmonic oscillator the angular frequency ω for oscillation is constant.

The solution of differential equation is $\frac{d^2x}{dt^2} + \omega^2 x = 0$ and $x = c_1 \cos \omega t + c_2 \sin \omega t$ the c_1 and c_2 can be determined by initial condition on position x and velocity $\dot{x} = \frac{dx}{dt}$.

The position $x = c_1 \cos \omega t + c_2 \sin \omega t$ can be also written as $x = A \cos(\omega t + \phi)$ where $c_1 = A \cos \phi$ and $c_2 = -A \sin \phi$.

The term A is identified as amplitude which is identified as maximum displacement and ϕ is initial phase. Initial phase will

$$A = \sqrt{c_1^2 + c_2^2} \text{ and } \phi = \tan^{-1} \left(-\frac{c_2}{c_1} \right)$$

The velocity can be given by $\dot{x} = \frac{dx}{dt} = -A\omega \sin(\omega t + \phi)$

But best solution of the $\frac{d^2x}{dt^2} + \omega^2 x = 0$ x is equal to Real part of function $A \exp i(\omega t + \phi)$.

We can write $x = \text{Re} [A \cos(\omega t + \phi) + i f(t)]$

If we choose the simplest $f(t)$ as $f(t) = A \sin(\omega t + \phi)$ then the locus between real part $x = A \cos(\omega t + \phi)$ and $y = A \sin(\omega t + \phi)$

The solution of simple harmonic oscillator is identified as projection of radius vector with magnitude A along x or real axis.

